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XGBoost-Based Susceptibility Model Exhibits High Accuracy and Robustness in Plateau Forest Fire Prediction

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Abstract

Forest fire susceptibility prediction is essential for effective management, yet considerable uncertainty persists under future climate change, especially in climate-sensitive plateau regions. This study integrated MODIS fire data with climatic, topographic, vegetation, and anthropogenic variables to construct an Extreme Gradient Boosting (XGBoost) model for the Yunnan Plateau, a region highly prone to forest fires. Compared with Support Vector Machine and Random Forest models, XGBoost showed superior ability to capture nonlinear relationships and delivered the best performance, achieving an AUC of 0.907 and an overall accuracy of 0.831. The trained model was applied to climate projections under SSP1-2.6, SSP2-4.5, and SSP5-8.5 to assess future fire susceptibility. Results indicated that high-susceptibility periods primarily occur in winter and spring, driven by minimum temperature, average temperature, and precipitation. High-susceptibility areas are concentrated in dry-hot valleys and mountain basins with elevated temperatures and dense human activity. Under future climate scenarios, both the probability and spatial extent of forest fires are projected to increase, with a marked expansion after 2050, especially under SSP5-8.5. Although the XGBoost model demonstrates strong generalizability for plateau regions, uncertainties remain due to static vegetation, coarse anthropogenic data, and differences among climate models.

Keywords: climate change scenarios; machine learning; forest fire susceptibility prediction; XGBoost; Yunnan Plateau; multi-source data integration

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1. Introduction

Uncontrolled forest fires severely damage ecosystem integrity, reducing carbon sequestration, changing habitat structure, weakening resilience, incurring substantial socioeconomic costs, and endangering public health through the release of hazardous substances [1,2].

In recent decades, anthropogenic greenhouse gas emissions have served as the primary driver of global climate change [3]. Specifically, these emissions have influenced global warming, altered precipitation patterns, and increased the frequency of extreme weather events [4,5]. These changes have greatly increased the frequency, duration, and severity of droughts globally, intensifying both the occurrence and severity of forest fires [6,7]. Climate-induced heatwaves and droughts decrease fuel moisture, lower ignition thresholds, and

enhance fire intensity and spread. As a result, the increased frequency and severity of extreme weather events have led to widespread vegetation damage and fuel accumulation [8], with global warming also increasing lightning activity and wildfire occurrence [9]. Therefore, understanding trends in forest fire susceptibility—defined here as the relative spatial tendency of fire occurrence under given environmental conditions—rather than the exact probability of an event at a specific time under future climate change scenarios has become a key research focus [10].

Understanding the responses of forest fires to climate change is essential for developing projection models and implementing mitigation strategies. Previous studies have assessed the probability of fire occurrence, defined as the modeled likelihood of fire events based on historical fire records and environmental predictors, as well as the intensity and extent of future fires under various climate scenarios [11]. However, several limitations remain in previous research. First, most studies relied on coarse-resolution climate data, failing to capture the fine-scale heterogeneity of complex mountainous terrain and local climatic variations [12–14]. Second, some studies adopted linear overlay or empirical weighting approaches [15,16], which, although revealing correlations between temperature, precipitation, drought, and fire activity, could not effectively capture nonlinear interactions among variables or quantify their relative importance. Third, while physically based models [17] offer theoretical value in elucidating the climatic control mechanisms of wildfires, their limited input variables often neglect vegetation and anthropogenic factors, constraining a comprehensive understanding of the multidimensional drivers of fire occurrence. However, future forest fire patterns remain uncertain due to regional variability in forest fire drivers such as climate, vegetation, topography, and human activity [18,19]. In regions where the climate is a key regulator for forest fires, future climate change often has a major impact on the fire patterns in that area. Therefore, gaining a deeper understanding of the patterns of fire susceptibility in future climate-sensitive areas is crucial for forest fire prevention and control.

Modeling the relationships between forest fires and their driving factors is considered crucial to understanding susceptibility [20]. Statistical and machine learning models [21,22], including XGBoost [23], RF [24], SVM [25], and gradient boosted decision trees (GBM) [26], have been widely used for this purpose. Considering the nonlinear, interactive nature of fire-driving mechanisms, machine learning techniques are well-suited to capture these complexities [27]. Among these, XGBoost, through its iterative boosting strategy, has demonstrated superior performance in modeling nonlinear relationships, handling high-dimensional and correlated features, and preventing overfitting via built-in regularization. In this model, climatic, topographic, vegetative, and anthropogenic factors jointly influence fire occurrence through complex nonlinear interactions. Climate change alters temperature, humidity, and precipitation patterns, affecting fuel drying rates [28–30]. Topography and wind regimes together determine the direction and intensity of fire spread [31], with human activities increasing ignition probability under drought conditions [32]. These interactions among climate, topography, vegetation, and human factors contribute to greater spatial and temporal uncertainty in fire distribution and frequency under future climate scenarios [5].

Despite the growing number of studies on large-scale fire susceptibility, localized assessments in climate-sensitive plateau regions remain scarce [33]. For example, the Yunnan Plateau in southwestern China is considered a global forest fire hotspot, characterized by complex topographic variability, pronounced climatic gradients, significant microclimatic changes, and climate response sensitivity, resulting in high spatial heterogeneity in fire-driving mechanisms [10,34,35]. However, the data used in the studies of large-scale forest fire susceptibility often have low spatial resolution, limiting the ability to capture local fire dynamics, and thereby making it difficult to fully reflect the spatial heterogene-

ity and response sensitivity of climate change-driven mechanisms at a local scale [36,37]. These coarse-resolution datasets tend to obscure local high-susceptibility zones, potentially underestimating fire hazards and constraining the precision of resource allocation and early-warning systems in fire management. The present study investigated the climate-sensitive Yunnan Plateau and employed the XGBoost model to establish relationships between fire occurrence and local climatic, topographic, vegetative, and anthropogenic factors. We aimed to identify key driving variables, evaluate spatial susceptibility under historical (2000–2020) and projected future climate scenarios (SSP1-2.6, SSP2-4.5, and SSP5-8.5), and the evolution of fire susceptibility from 2020 to 2100. The objectives were (1) to identify historical forest fire hotspots; (2) determine key factors influencing forest fire occurrence; and (3) assess potential spatiotemporal trends in fire susceptibility under future climate scenarios. This study also provided recommendations for optimizing forest fire management strategies in climatically heterogeneous plateau regions under future climatic conditions.

2. Materials and Methods

2.1. Study Site Description

Yunnan Plateau, located in southwestern China (21°08′–29°15′ N, 97°31′–106°11′ E), is positioned on the southeastern edge of the Qinghai–Tibet Plateau (Figure 1). It borders the Bay of Bengal and the South China Sea to the south, with a complex terrain and numerous valleys, ranging in elevation from 76 m to over 6100 m, covering approximately 394,000 km^2. Characterized by a subtropical monsoon climate, the Yunnan Plateau experiences distinct wet and dry seasons, with substantial spatial variation in temperature and precipitation. Annual rainfall averages 1100 mm, but varies widely across the province, with over 80% of total annual precipitation occurring from May to October. The mean annual temperature ranges from 5 °C to 24 °C. Due to its unique geographical location, the Yunnan Plateau is particularly climate-sensitive. Over the past few decades, the Yunnan Plateau has exhibited a pronounced warming and drying trend, accompanied by increasing extreme climate events [38,39], such as droughts. As a global biodiversity hotspot, the Yunnan Plateau hosts diverse forest ecosystems, including subtropical evergreen broadleaf, coniferous, and mixed forests that cover approximately 53% of the land area [20,39]. Among these, coniferous forests are predominant in the central mountainous areas and northwestern Yunnan, with high flammability and thick litter layers making them easily ignitable during the dry season [40]. Human activity is mainly concentrated in the central basin urban agglomerations and major river valleys, where agricultural burning and traditional ritual fires remain frequent, increasing potential ignition sources.

These associations among climate, topography, vegetation, and human activity result in pronounced heterogeneity among fire-driving mechanisms across the Yunnan Plateau. For example, low-elevation, dry, and hot river valleys, such as the Nujiang and Yuanjiang basins, are prone to frequent wildfires due to limited rainfall, strong evaporation, and dry combustible materials. By contrast, high-altitude areas, with lower temperatures and higher humidity, experience lower wildfire frequencies. Compared to adjacent regions such as the hilly areas of South China, the Yunnan plateau experiences more intense drying of combustible materials during the winter and spring, leading to significantly higher fire frequency [41,42]. Therefore, the Yunnan plateau can be regarded as a model region for investigating forest fire risk under current and future climate conditions.

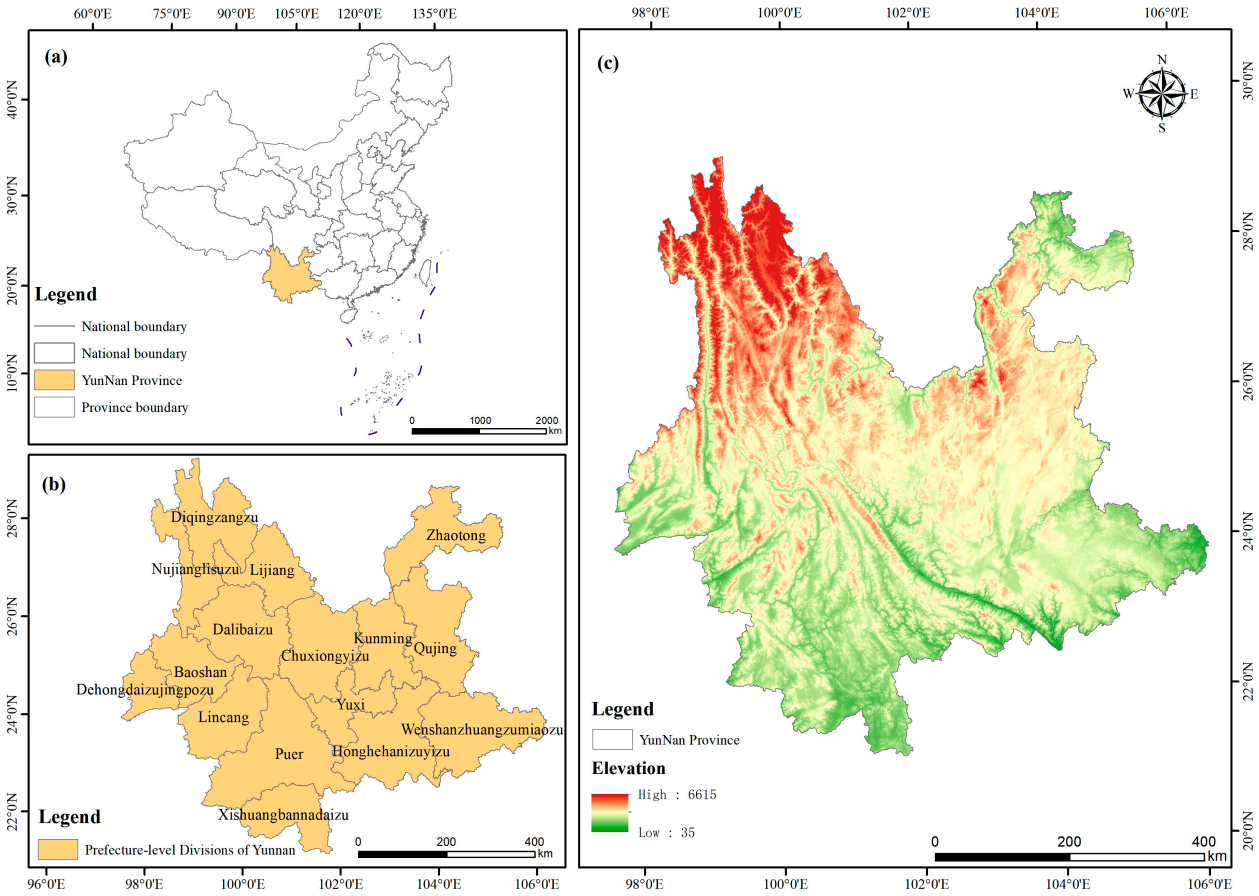


Figure 1. Location of the study area in the Yunnan Plateau, China. (**a**) Geographic location of Yunnan in China; (**b**) Prefecture-level administrative divisions of Yunnan Province; (**c**) Spatial distribution of elevation across the Yunnan Plateau.

2.2. Data

2.2.1. Forest Fire Data

This study used the MODIS active fire product (MCD14ML, Collection 6.1) provided by NASA Fire Information for Resource Management System (FIRMS), National Aeronautics and Space Administration (NASA), Greenbelt, MD, USA. Since November 2000, this dataset has recorded daily global observations of thermal anomalies at a 1 km resolution, including the latitude, longitude, observation time, brightness temperature, fire radiative power (FRP), and confidence level, making it a widely used resource for fire research at regional and global scales [43,44].

The MODIS data from 2000 to 2020 were preprocessed using ArcGIS 10.8, and we selected fire points that occurred during critical fire prevention periods. Only fire points with ≥80% detection confidence were used for analysis. To eliminate redundancy, fire points within 3 days and 3 km were merged and represented by the first detected point. In addition, fire points outside forest areas were excluded using the 2020 GLC_FCS30 land cover datasets. After filtering, 9154 valid forest fire points were identified. Non-fire samples, equal in number to fire samples, were uniformly generated beyond a 3 km buffer from all fire points to represent background conditions without fire occurrence. As shown in Figures 2 and 3, 98.4% of forest fires in Yunnan occurred during the critical fire prevention periods, with a peak from January to May. Annually, fire frequency fluctuated, peaking in 2010 during a major drought from late 2009 to early 2010 [45]. The fire points were predominantly located in the northwestern and southern regions, aligning with previous reports, thus validating the spatiotemporal fire data.

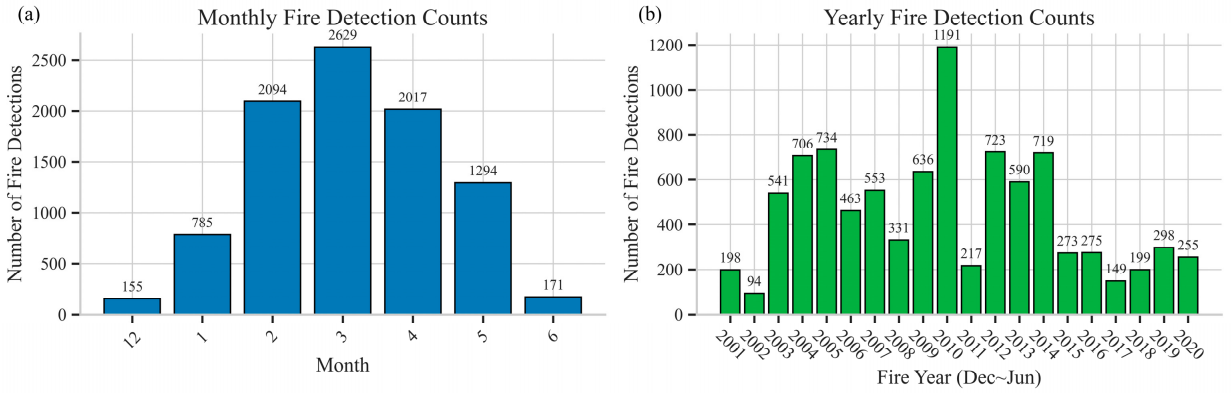


Figure 2. Monthly and yearly distributions of forest fire detections during key fire prevention periods (2000–2020) (**a**,**b**).

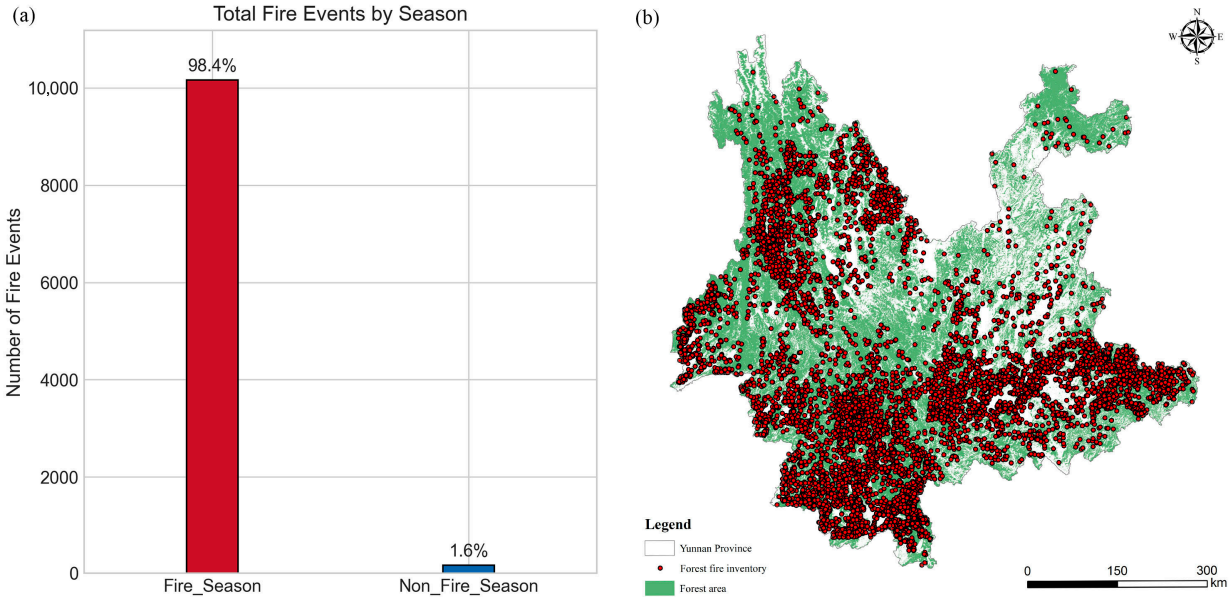


Figure 3. Fire and non-fire prevention period distribution of forest fire events (2000–2020) (**a**); spatial distribution of forest fire events in the Yunnan Plateau during 2000–2020 (**b**).

2.2.2. Predictor Variables

In this study, we selected 11 predictor variables, which were categorized into climatic (maximum, minimum, and average temperatures, VPD, precipitation, wind speed, relative humidity), vegetative (NDVI, vegetation type), topographic (elevation), and anthropogenic (distance to built-up areas) (Figure 4). The selection of variables was based on previous studies on forest fire susceptibility modeling [19,27,34,46–48] and further optimized according to the regional characteristics of the Yunnan Plateau. Unlike studies that used only NDVI to represent vegetation coverage, the present study introduced the vegetation type variable to characterize differences in fuel structure and composition. Moreover, for anthropogenic factors, the distance to built-up areas was used instead of the traditional distance to roads, as it more accurately reflected the spatial influence of human settlement and ignition activities. Elevation data were obtained from NASA's global Shuttle Radar Topography Mission Digital Elevation Model (SRTM DEM), NASA, USA, while NDVI was extracted from the MODIS vegetation index product (MOD13A2), NASA, USA. Using multi-year mean values for the fire season (1 December–15 June, 2001–2020) to represent long-term vegetation cover and fuel conditions. The vegetation type data were derived from the 2020 GLC_FCS30 forest classification dataset, developed by the Aerospace Information Research

Institute, Chinese Academy of Sciences, Beijing, China, using three forest categories: evergreen needleleaf, evergreen broadleaf, and deciduous broadleaf forests. Built-up area data sourced from ESA's WorldCover 10 m 2020 global land cover product, Paris, France, based on Sentinel-1 and Sentinel-2 imagery, with 10 m resolution and 11 land cover classes. Built-up areas were identified using land cover code 50 to indicate human influence on fire risk, and the distance to built-up areas variable was calculated using the Euclidean distance method to represent the spatial proximity effect of human ignition sources. Distance classes were generated using the natural breaks (Jenks) classification method, which minimizes within-class variance while maximizing between-class differences in distance values. Current climate variables (maximum, minimum temperature, VPD, precipitation, and wind speed) were derived from the TerraClimate monthly dataset. The relative humidity and average temperature were extracted from the ERA5 daily reanalysis dataset provided by ECMWF (Table 1) [49]. All climate variables were based on the multi-year averages from 1984 to 2014 to characterize the long-term climatic baseline. Daily 2 m air temperature and dew point temperature were used to compute relative humidity using the Tetens formula:

$$e_s(T) = 0.6108 \times exp\left(\frac{17.27T}{T+237.3}\right), \quad (1)$$

$$e(T_d) = 0.6108 \times exp\left(\frac{17.27T_d}{T_d+237.3}\right), \quad (2)$$

$$RH = \left(\frac{e(T_d)}{e_s(T)} \times 100\%\right), \quad (3)$$

where T is the daily mean air temperature, T_d is the daily mean dew point temperature in (°C), and $e_s(T)$ and $e(T_d)$ represent the saturation vapor pressure and actual vapor pressure, respectively (kPa).

Table 1. Data description of forest fire influence.

No.	Data	Scale/Resolution	Unit	Original Data Format	Source
1	Average temperature	0.25° (lat/long)	°C	NetCDF	ERA5
2	Relative humidity	0.25° (lat/long)	%	NetCDF	ERA5
3	Maximum temperature	0.041° (lat/long)	°C	NetCDF	TerraClimate
4	Minimum temperature	0.041° (lat/long)	°C	NetCDF	TerraClimate
5	Precipitation	0.041° (lat/long)	mm	NetCDF	TerraClimate
6	Vapor pressure deficit	0.041° (lat/long)	kpa	NetCDF	TerraClimate
7	WindSpeed	0.041° (lat/long)	m/s	NetCDF	TerraClimate
8	Elevation	90 m	m	Raster	SRTM
9	NDVI	1 km	ratio	Raster	MODIS
10	Vegetation type	30 m	-	Raster	GLC_FCS30
11	Distance to built	-	km	Vector	ESA WorldCover
12	Forest fire points	1 km	-	Vector	MODIS

Future climate data under various carbon emission scenarios were obtained from the CMIP6 project [50], specifically the BCC-CSM2-MR model under three scenarios of SSP1-2.6, SSP2-4.5, and SSP5-8.5 [51]. Previous studies demonstrated this model's ability to simulate China's future climate patterns [52]. The present study applied the delta method [53] for bias correction, using the multi-year mean (1984–2014) differences between observations and historical simulations to calculate the spatial bias fields for each climatic variable, which were used to adjust projected climate data from 2020 to 2100. It should be noted that the downscaling of CMIP6 climate projections from a native resolution of approximately 100 km to 1 km using the delta method and bilinear interpolation does not explicitly resolve fine-scale topographic effects, such as valley–slope contrasts or elevation-dependent microclimatic variability. Instead, this approach preserves large-scale climate

change signals simulated by the global climate model while transferring them onto high-resolution observational climatologies that already incorporate local topographic controls. Therefore, the resulting high-resolution climate fields are suitable for capturing regional climate gradients and relative spatial patterns of change, but they may underestimate localized microclimatic heterogeneity in complex mountainous terrain. All variables were resampled to a 1 × 1 km resolution using bilinear interpolation in ArcGIS, and raster data were extracted based on the Yunnan Plateau boundary.

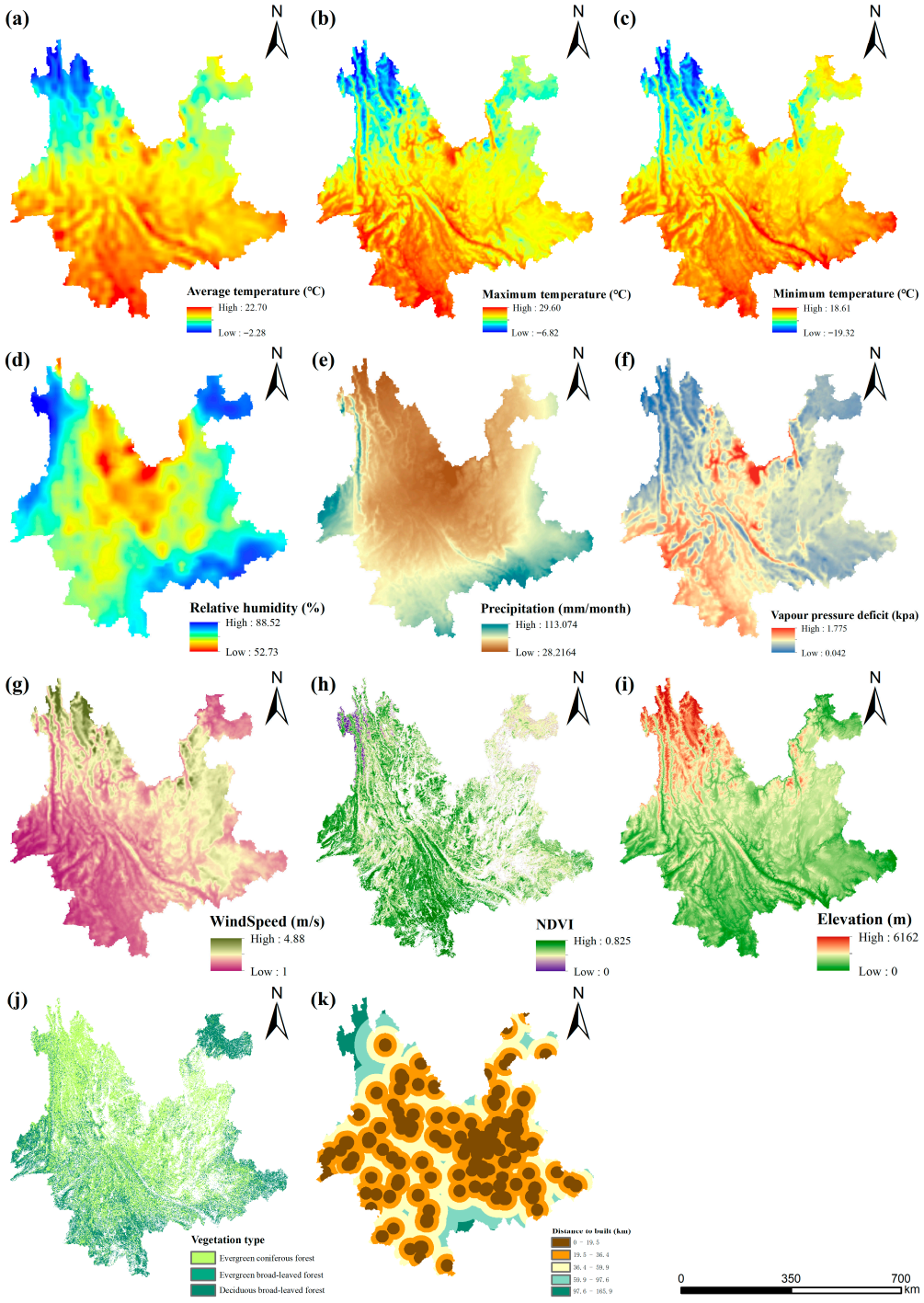


Figure 4. Forest fire conditioning factors. (**a**–**h**) Represent multi-year mean conditions during the fire season (1 December to 15 June) for the period 2001–2020, including climatic and vegetation-related variables. (**i**–**k**) Represent single-year variables for 2020, including elevation, vegetation type, and distance to built-up areas.

2.3. Technical Workflow

The workflow of this study comprised four stages: sample construction, variable extraction, model training and evaluation, and forest fire susceptibility prediction (Figure 5).

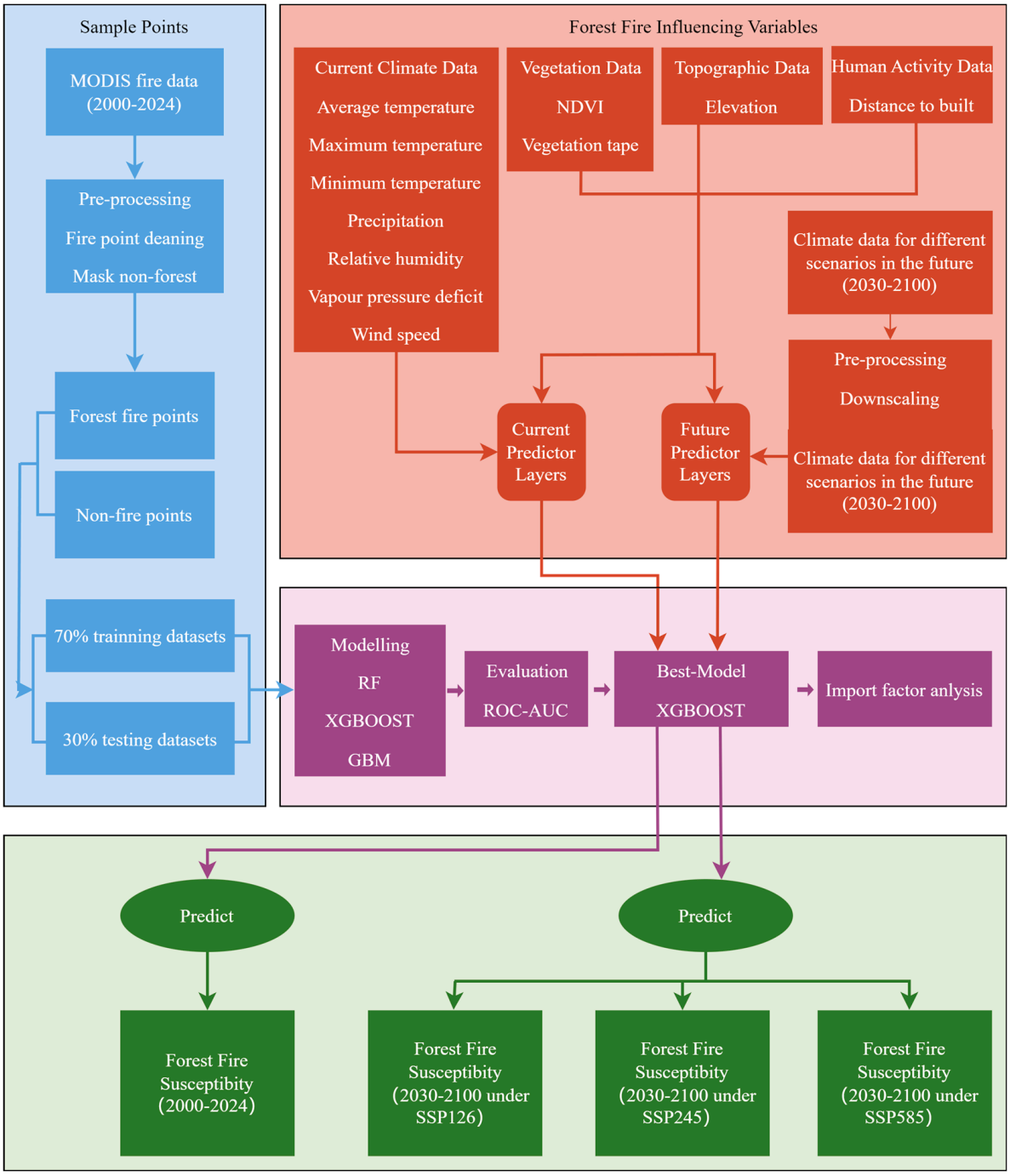


Figure 5. Technology workflow in this study.

In the first stage, fire samples were derived from MODIS active fire products (2000–2020) after strict filtering, retaining only high-confidence fire events occurring during the key fire-prevention period. An equal number of non-fire samples was randomly generated in forested zones beyond the fire buffer areas to represent fire-safe regions, ensuring a balanced number of fire and non-fire samples. The complete dataset was then partitioned into training (70%) and testing (30%) subsets using a spatially independent hold-out strategy.

In the second stage, 11 predictor variables, including average temperature, maximum temperature, minimum temperature, precipitation, wind speed, relative humidity, vapor pressure deficit (VPD), NDVI, vegetation type, elevation, and distance to built, were preprocessed via temporal filtering and resampled to a 1 km spatial resolution. Future predictor layers were selected from BCC-CSM2-MR climate simulations under SSP1-2.6, SSP2-4.5, and SSP5-8.5 scenarios. These datasets were bias corrected and downscaled.

In the third stage, current and future predictors, along with labeled sample points, were used to train three machine learning algorithms, namely, RF, XGBoost, and GBM. Model performance was evaluated using ROC-AUC, F1 score, accuracy, precision, and recall. Among the evaluated machine learning algorithms, XGBoost was further used for feature importance analysis.

In the final stage, the trained XGBoost model was used to simulate forest fire susceptibility under current (2000–2020) and future climate conditions (2020–2050, 2050–2080, and 2080–2100). The model output represented the probability of fire occurrence, and spatial variations in future fire susceptibility were analyzed to identify potential shifts in high-susceptibility zones under changing climate scenarios.

2.4. Machine Learning Model Application

Prior to forest fire susceptibility prediction, the machine learning models were parameterized through hyperparameter tuning. These hyperparameters control model complexity and learning behavior, including the number of trees, maximum tree depth, rate of learning, subsampling, variable subsetting, and regularization parameters to prevent overfitting [23]. Hyperparameters of the XGBoost, RF, and SVM models were optimized using a Cartesian grid search, in which commonly used parameter ranges were systematically explored. The optimal hyperparameter combination for each model was selected based on predictive performance during training. To avoid potential overestimation of model performance caused by spatial autocorrelation, model training and evaluation were conducted using spatially independent data partitions. Specifically, spatial blocking was applied to group samples into spatial units, and spatial cross-validation was implemented to ensure that training and validation samples were geographically separated. Model performance was therefore assessed under spatially independent conditions rather than simple random partitioning.

2.4.1. XGBoost

XGBoost is an optimized implementation of the gradient boosting algorithm that improves predictive performance and computational efficiency through parallel tree construction and regularization strategies [23,54]. It builds an ensemble of decision trees sequentially, where each new tree is trained to correct the residuals of the previous ensemble, enabling the model to capture complex nonlinear relationships between predictors and fire occurrence.

2.4.2. RF

RF is an ensemble learning method that constructs a large number of decision trees using bootstrap samples of the training data and random subsets of predictor variables, and aggregates individual tree predictions through majority voting or averaging [24]. This structure enables RF to effectively model nonlinear relationships while reducing overfitting caused by individual trees.

2.4.3. SVM

SVM is a supervised learning algorithm grounded in statistical learning theory. SVM aims to identify an optimal separating hyperplane that maximizes the margin between classes, thereby minimizing the generalization error. To address nonlinear classification

problems, SVM employs kernel functions to project input data into a higher-dimensional feature space [55]. In this study, an SVM with a radial basis function (RBF) kernel was implemented. The penalty parameter (C), which controls the trade-off between margin width and misclassification error, and the kernel parameter (gamma), which determines the influence range of individual samples, were optimized using a Cartesian grid search. Hyperparameter selection was guided by spatial cross-validation to improve model generalization and reduce spatial dependency effects.

2.4.4. Variable Importance

Variable importance was assessed to quantify the relative contribution of each predictor to forest fire susceptibility, addressing the interpretability of model outputs [56]. For tree-based models, variable importance was derived from the internal model structure, based on the contribution of each predictor to node splitting and impurity reduction during model training.

2.4.5. SHAP Summary Plot

SHAP (SHapley Additive exPlanations) is an explainable artificial intelligence method rooted in cooperative game theory, which assigns each predictor a Shapley value representing its marginal contribution to an individual prediction [57,58]. To further enhance model interpretability, SHAP was applied to the XGBoost model. This framework ensures a consistent and theoretically grounded attribution of feature effects. The SHAP summary plot provides a global overview of model behavior by visualizing both the importance and direction of influence of each predictor across all samples. Features with larger absolute SHAP values exert stronger impacts on fire susceptibility predictions, while the color gradient indicates whether high or low feature values increase or decrease predicted fire risk. This visualization enables an intuitive understanding of how climatic, environmental, and anthropogenic factors jointly influence forest fire occurrence.

3. Results

3.1. Performance of the Models

The XGBoost model outperformed the other models across all indicators, achieving an AUC of 0.907, indicating excellent discrimination between the fire and non-fire point samples (Table 2). The accuracy, precision, recall, and F1 score were 0.831, 0.877, 0.792, and 0.832, respectively, the highest among the three models. A Kappa coefficient of 0.662 also demonstrated strong agreement between predictions and actual observations. By contrast, SVM exhibited the weakest performance, with an AUC of 0.753 and an F1 score of 0.714. Although RF performed better than SVM, it remained less effective than XGBoost. The ROC curves further supported these findings, where XGBoost maintained consistently better performance, especially at lower false positive rates, as indicated by the largest AUC area (Figure 6).

Table 2. Comparison of prediction performance metrics across models.

Model	AUC	Accuracy	Precision	Recall	F1 Score	Kappa	Sensitivity
SVM	0.753	0.716	0.718	0.71	0.714	0.431	0.71
RF	0.894	0.814	0.865	0.769	0.814	0.629	0.809
XGBoost	0.907	0.831	0.877	0.792	0.832	0.662	0.817

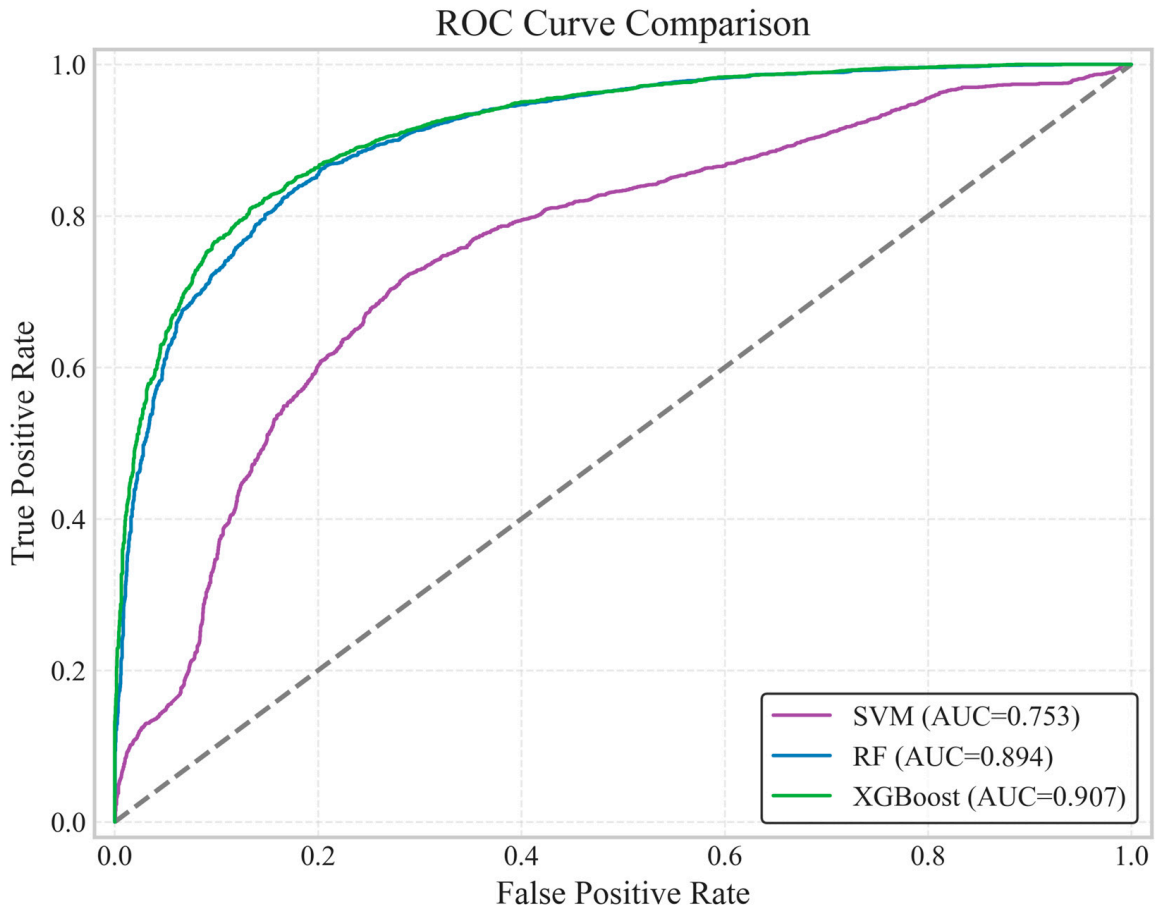


Figure 6. ROC curve comparison of fire susceptibility models.

Forest fire susceptibility maps for Yunnan from 2000–2020 were generated by SVM, RF, and XGBoost (Figure 7). This study classified the model-predicted fire probabilities into four risk levels, where less than 25%, 25–50%, 50–75%, and greater than 75% corresponded to low, moderate, high, and very high risk zones, respectively. Although the three models demonstrated consistency in identifying high-susceptibility areas, they differed significantly in spatial continuity and classification granularity, especially in central-southern Yunnan. The XGBoost model produced a more distinct spatial structure and clearer risk gradients. High (50–75%) and very high (≥75%) risk zones were more extensive and spatially coherent, mainly predominantly located in southeastern, southwestern, and northwestern mountainous regions, aligning with the historical fire distribution shown in Figure 3b. Although RF identified similar hotspots, its medium-susceptibility zones were fragmented. By comparison, SVM yielded more conservative predictions, assigning most areas to low (≤25%) or moderate (25–50%) risk classes and detecting few high-susceptibility zones, reflecting limited sensitivity to complex terrain fire dynamics.

Overall, the combined model evaluation metrics and spatial results confirmed that XGBoost provided superior spatial mapping and risk prediction continuity. Accordingly, XGBoost was selected as the final model for future forest fire susceptibility projections and scenario mapping in Yunnan, due to its robustness and performance.

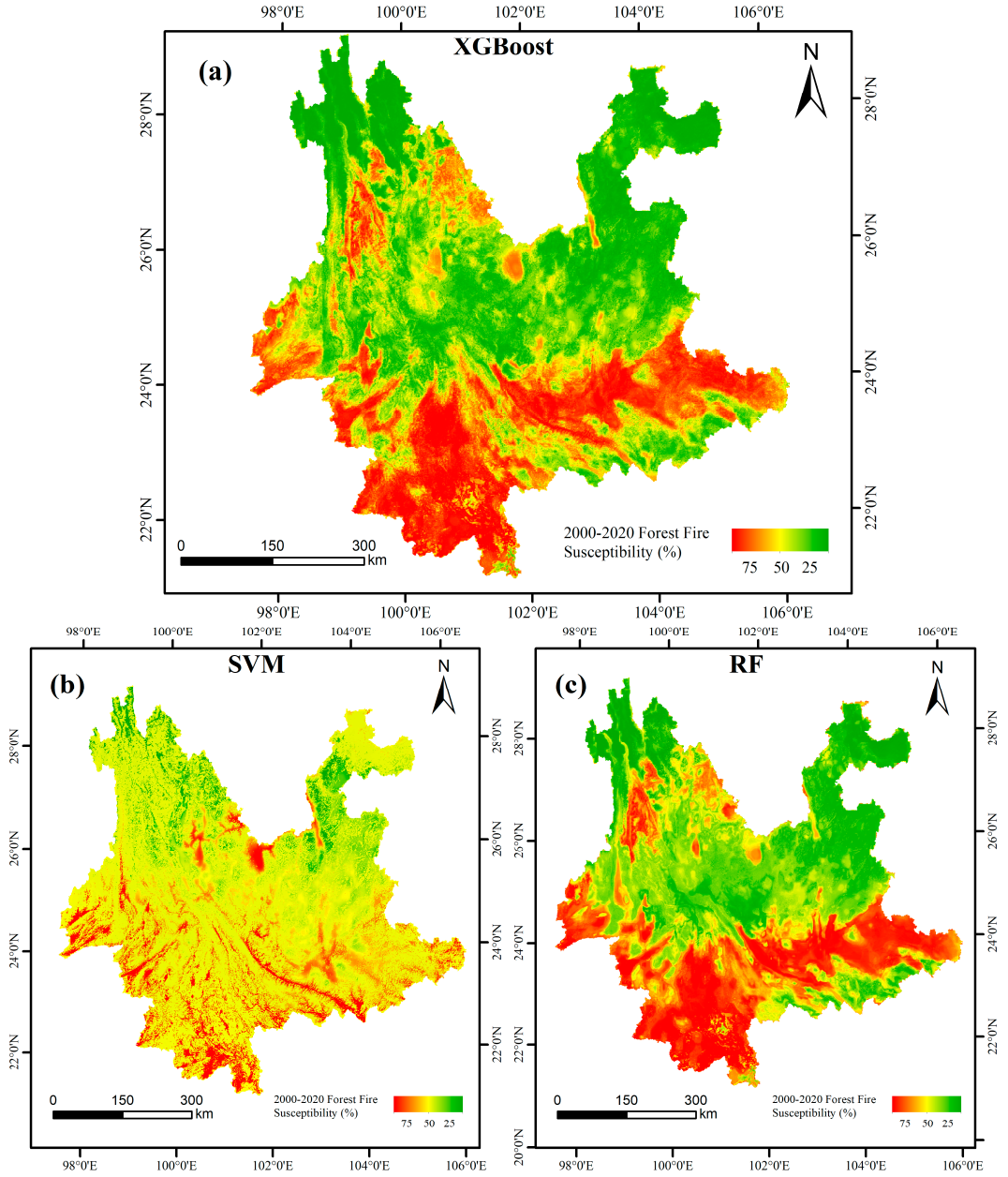


Figure 7. Forest fire susceptibility mapping in the Yunnan Plateau using different machine learning models (2000–2020).

3.2. Importance of Predictor Variables

Hydrothermal climate variables were critical in fire susceptibility during Yunnan's high fire season, followed by topographic, vegetative, and anthropogenic factors (Figures 8 and 9). The minimum temperature was the most influential variable (importance = 0.175), followed by average temperature (0.145), precipitation (0.105), maximum temperature (0.104), and relative humidity (0.091). Together, these five variables explained approximately 62% of the model's variance, underscoring the dominant role of climate in fire dynamics.

The SHAP further clarified each variable's effect, where each point represented an observation, with red indicating high and blue indicating low variable values. Higher minimum and average temperatures were associated with increased SHAP values, suggesting greater fire susceptibility. Elevated nighttime temperatures reduced cooling, prolonged warm conditions, and accelerated fuel drying, increasing the likelihood of ignition. Simi-

larly, lower precipitation was associated with high SHAP values, suggesting that drought reduced fuel moisture and enhanced flammability. Relative humidity and VPD represented atmospheric moisture and dryness, respectively, while low humidity and high VPD significantly elevated fire risk.

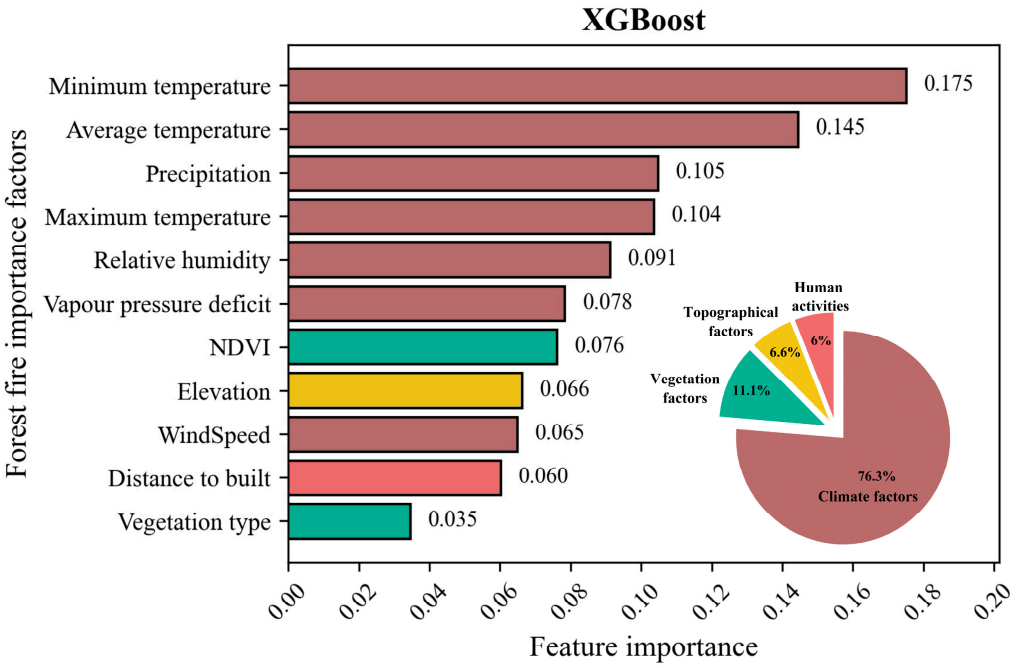


Figure 8. The importance of conditioning factors.

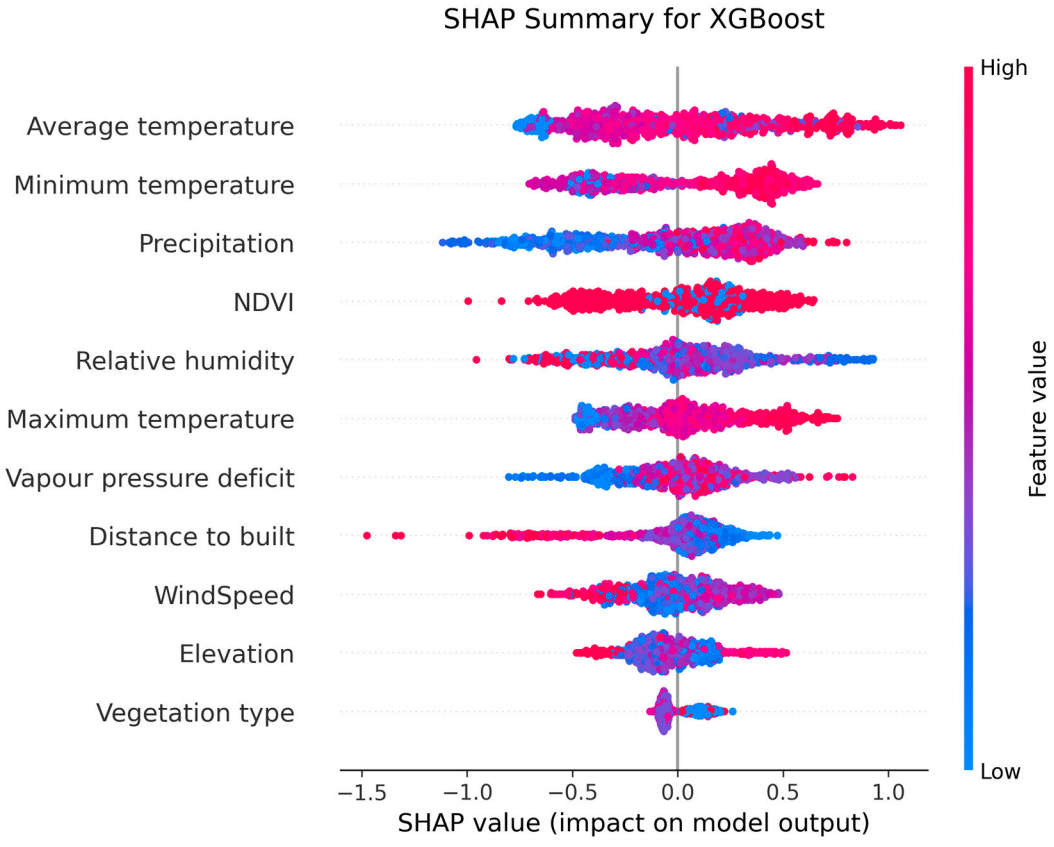


Figure 9. SHAP summary plot of predictor variables in the XGBoost model.

Among static variables, NDVI (0.076) and elevation (0.066) were significant. The SHAP values suggested that very low or very high NDVI was associated with lower fire risk, while moderate NDVI possibly indicated abundant but dry biomass, increasing susceptibility. Wind speed (0.065) had a moderate effect, likely indicating its influence on fire spread. Distance to built-up areas (0.060) was negatively correlated. Notably, fires were more likely near settlements, due to agricultural burning, land clearing, and cultural practices such as Qingming Festival rituals in the rural-urban mosaic of Yunnan [33].

In summary, both the feature importance and SHAP results confirmed that temperature and moisture variables dominated forest fire dynamics in Yunnan, especially during critical fire prevention periods. These findings highlighted the need to enhance fire monitoring and prevention in climate-sensitive and anthropogenically affected zones under ongoing global warming.

3.3. Forest Fire Susceptibility Under Current and Future Scenarios

Overall, fire susceptibility increased over time, with more significant growth under high-emission scenarios (Table 3). Using 2000–2020 as the baseline, the mean fire probability was 0.452, and high-susceptibility zones (probability > 0.5) covered 41.6% of forested areas. Under the low-emission SSP1-2.6 scenario, future changes were minimal, where mean regional fire probability remained between 0.42 and 0.46, and the proportion of high-susceptibility areas increased slightly from 0.423 (2020–2050) to 0.456 (2050–2080), an increase of 7.8%. A slight decline to 0.449 was observed by 2080–2100, suggesting that stringent mitigation policies possibly effectively stabilized fire risk.

Table 3. Summary statistics of forest fire susceptibility.

Scenario	Period	Mean	Max	Min	Std	HighRiskPct
Present	2000–2020	0.452	0.985	0.010	0.307	0.416
SSP1-2.6	2020–2050	0.423	0.956	0.014	0.234	0.354
SSP1-2.6	2050–2080	0.456	0.956	0.014	0.231	0.427
SSP1-2.6	2080–2100	0.449	0.945	0.014	0.221	0.421
SSP2-4.5	2020–2050	0.430	0.958	0.014	0.233	0.370
SSP2-4.5	2050–2080	0.468	0.958	0.013	0.237	0.428
SSP2-4.5	2080–2100	0.490	0.942	0.014	0.232	0.477
SSP5-8.5	2020–2050	0.445	0.964	0.014	0.228	0.408
SSP5-8.5	2050–2080	0.518	0.944	0.013	0.210	0.548
SSP5-8.5	2080–2100	0.522	0.942	0.073	0.158	0.533

Under the medium-emission SSP2-4.5 scenario, fire probability increased, from 0.453 (2020–2050) to 0.490 (2080–2100), representing a net rise of 8.2%. The proportion of high-susceptibility areas also rose from 42.0% to 47.7%, suggesting that even under moderate mitigation efforts, fire risk continued to escalate. In contrast, the high-emission SSP5-8.5 scenario demonstrated the most pronounced increase, with average probability rising from 0.445 in the baseline period to 0.522 in 2080–2100, a 17.3% increase. The corresponding high-susceptibility area expanded to 54.8% by 2050–2080, indicating that unabated emissions possibly substantially amplified fire susceptibility.

Forest fire susceptibility in Yunnan under prevailing climate conditions (2000–2020) exhibited a pronounced spatial heterogeneity, with the map in the figure utilizing a continuous color gradient to represent fire occurrence probability (Figure 7), with green areas indicating low fire risk and red areas indicating high-susceptibility zones. Forest fire susceptibility was uneven across the province, with high-susceptibility areas primarily concentrated in the southern, southeastern, and northwestern regions, especially in Pu'er, Xishuangbanna, and Honghe. These regions were located in low-latitude zones with hot

and dry climates, where winter-spring seasons are typically dominated by the East Asian winter monsoons, leading to low precipitation, reduced relative humidity, and elevated minimum temperatures. Combined with dense forest coverage and frequent human disturbances (i.e., agricultural burning, land clearing), these factors jointly drove high fire susceptibility [59].

Northwestern areas, including western Dali, northern Chuxiong, and the mid-to-lower reaches of the Nujiang River, presented higher fire susceptibility. Although located at higher latitudes, these regions are dominated by hot-dry valleys and mountainous basins, experiencing intense surface evapotranspiration and poor precipitation conditions. Elevated vapor pressure deficits (VPD), coupled with low humidity and high temperatures, render these areas especially vulnerable during drought years, forming a climatic context highly conducive to fire occurrence.

In contrast, northern and western high-altitude regions such as Lijiang, Zhaotong, and Diqing displayed lower fire susceptibility. These areas are characterized by high elevations, cooler temperatures, abundant atmospheric moisture, and complex topography that naturally limits fire spread and function as physical firebreaks. Moreover, vegetation type and density, accessibility, and the extent of human disturbance were found to further influence the spatial pattern of fire susceptibility.

Overall, the current spatial distribution of forest fire susceptibility reflected the combined influence of climate drivers, anthropogenic disturbances, and topographic-ecological constraints. This highlighted the high sensitivity of forest fire occurrence to microclimatic variations within Yunnan's complex geographical setting.

Future projections (Figure 10) under high-emission scenarios indicated expansion in fire-prone areas, especially in southwestern (Xishuangbanna, Pu'er, Lincang), southeastern (Honghe, Wenshan), and central hilly regions (Yuxi, southern Chuxiong, Yuanjiang River basin). The largest increases were observed in northeastern Yunnan (Qujing, Zhaotong), central Yunnan (Kunming, Dali), and southeastern areas. Some regions, such as northern Yuxi, parts of Honghe and Wenshan, and the Hengduan Mountains, could experience reductions in fire probabilities.

From 2020 to 2050, fire hotspots expanded into Chuxiong, Yuxi, and southern Dali. Several areas transition from moderate to high risk, reflecting the intensifying effects of warming and drought. By 2050–2080, under SSP2-4.5 and SSP5-8.5, high-susceptibility zones merged into broader contiguous belts across Baoshan, Wenshan, and the Ailao-Wuliang Mountains. This clustering pattern reflected the combined effect of rising temperatures, declining humidity, and stronger winds accelerating fire propagation. By 2080–2100, fire susceptibility peaked under SSP5-8.5, with Pu'er, Lincang, and Xishuangbanna largely encompassed in high-susceptibility zones. By contrast, SSP1-2.6 remained relatively stable, with only localized increases in southwestern Dali, central Yuxi, and parts of Honghe, suggesting that low-emission policies could effectively suppress fire spread in the long term.

Under SSP2-4.5 and SSP5-8.5, increases exceeding +30% were observed in Zhaotong, southern Chuxiong, northern Lincang, southern Wenshan and Honghe, Kunming, and Qujing (Figure 11). In some cases (southern Chuxiong, northern Lincang, southern Wenshan), increases exceeded +50%, forming distinct fire risk hotspots. In contrast, areas such as northern Honghe and Yuxi, the Hengduan Mountains, and southern Pu'er and Xishuangbanna showed probability decreases below −10%, with some local reductions exceeding −30%.

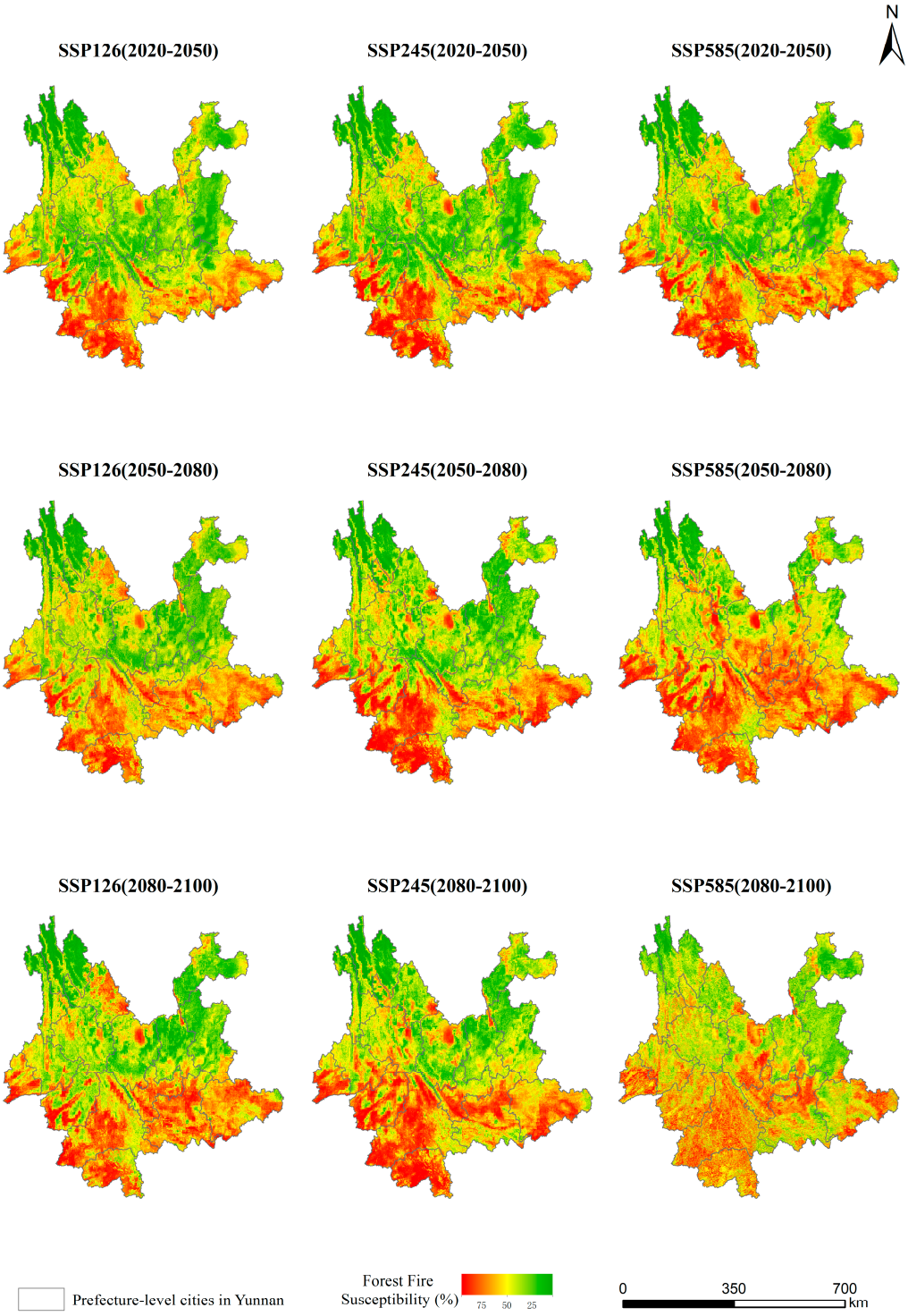


Figure 10. Prediction of forest fire occurrence in the Yunnan Plateau under BCC-CSM2-MR scenarios from 2020 to 2100.

Overall, Yunnan's future fire susceptibility followed a southern intensification, central-western expansion, and northern emergence pattern. This trend was most pronounced under SSP5-8.5, evolving from fragmented hotspots to expansive high-susceptibility zones. These dynamics could pose major threats to ecosystems and resource management, requiring the early deployment of adaptive fire infrastructure and zoned warning systems based on probability trends to achieve precision management goals.

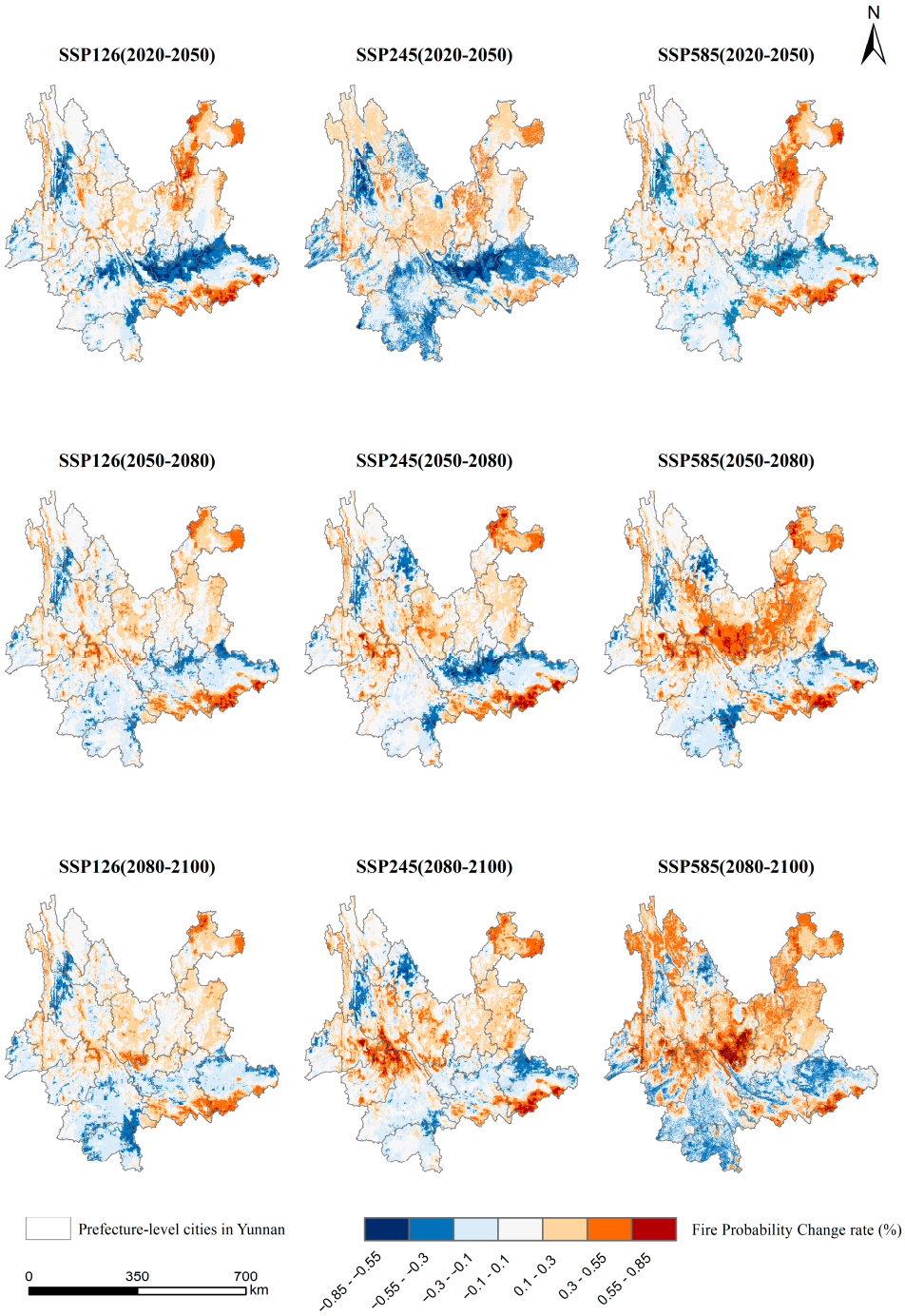


Figure 11. Spatial change rate of forest fire susceptibility from 2020 to 2100 under different SSP scenarios in the Yunnan Plateau.

4. Discussion

Global warming and the increasing frequency of drought events have increased forest fire risks, threatening ecosystem stability and socioeconomic security. As one of the most fire-prone regions in China, Yunnan has experienced frequent forest fires that negatively impact natural forests and agricultural lands. This study employed the XGBoost model, integrating MODIS fire data (2000–2020) and multi-source climatic, topographic, vegetative, and anthropogenic variables, to systematically assess the spatiotemporal distribution of current fire susceptibility and project future susceptibility (2020–2100) under different emission scenarios. The results revealed that minimum and average temperature were the most important predictors, followed by precipitation, relative humidity, and other

meteorological factors. Rising minimum and average temperatures not only accelerated fuel drying but also inhibited humidity recovery during traditionally low-susceptibility periods, such as nighttime or early morning. These findings also have direct implications for fire monitoring and preparedness. As nighttime and early-morning periods may no longer provide effective fuel moisture recovery under continued warming, fire surveillance strategies should be adjusted to extend monitoring beyond traditionally high-risk daytime windows. Specifically, nighttime patrols, automated infrared or thermal camera systems, and early-warning alerts based on minimum temperature and humidity thresholds could help detect and suppress ignitions that occur during these previously low-risk periods. In addition, adjusting crew readiness and resource allocation to account for elevated nocturnal fire risk may improve rapid response capacity under warming conditions. This extended the time window for fire ignition and increased overall susceptibility. Under future climate scenarios, increasing drought frequency and fuel accumulation may synergistically act to amplify potential fire risk [10]. This climate-fuel coupling mechanism may lead to longer fire durations and the wider spread of burned areas [2]. In addition, warming could alter wind patterns, increasing fire spread potential. These findings aligned with existing research, which demonstrated that temperature, relative humidity, and wind collectively drove fire occurrence [60–62].

The XGBoost model demonstrated superior predictive performance compared with RF and SVM. These findings were consistent with those of Hang et al. [48], who identified XGBoost as the optimal model in multi-model comparisons. Under current climate conditions, high-susceptibility forest fire zones in Yunnan were mainly concentrated in the south, southeast, and northwest, consistent with the findings reported by Shao et al. (2023) [63] using a CNN-based national fire susceptibility assessment for 2002–2010. Compared with deep learning models, which often lack interpretability, XGBoost provided better insights into variable contributions.

Future fire susceptibility projections under all three SSP scenarios show spatial expansion trends, intensifying in the south, shifting westward, and expanding across the central-west. Under the high-emission SSP5-8.5 scenario, this transition was pronounced by 2100, with high-susceptibility zones merging from localized hotspots into continuous regions. This highlighted the shaping influence of Yunnan's complex terrain on microclimates and fuel spatial patterns. Horn et al. (2025) [46] similarly noted that in eastern Germany, warming and decreased precipitation would jointly drive rising fire susceptibility, where solar radiation and human population density were key cofactors. By contrast, the present study found that although proximity to built-up areas was statistically significant, its effect size was overshadowed by climatic variables. This underscored the influence of climatic drivers on fire dynamics in Yunnan's diverse topography, suggesting regional variations in the interaction intensity between natural and anthropogenic factors [46].

Unlike Shao et al. (2023) [63], which assessed fire susceptibility at a national scale, the present study used a province-scale approach with high-resolution fire point and climate data, enhancing the detection of transitional zones (e.g., Yuxi, Baoshan). This approach improved spatial precision and sensitivity for local susceptibility assessment. Notably, we used distance to built-up areas as a proxy for human activity, instead of traditional measures such as distance to roads or city centers. This metric better reflected actual human disturbance ranges in Yunnan's rural-urban interface. Nevertheless, forest fire intensity and spread were governed by complex nonlinear interactions among weather, vegetation, and human factors [64], leading to uncertainty in future projections. In addition to uncertainties associated with climate model structure and emission scenarios, NDVI, vegetation type, and built-up areas were treated as temporally static variables in the future projections. This assumption does not account for potential changes in vegetation productivity, fuel struc-

ture, land-use conversion, or urban expansion under future climatic and socioeconomic development. Consequently, future fire susceptibility may be underestimated in regions where fuel accumulation intensifies, or overestimated in areas where land management or urbanization reduces fuel availability. Future studies should integrate dynamic vegetation models and time-evolving anthropogenic datasets to further refine fire susceptibility projections under changing environmental and socioeconomic conditions.

Although tree-based models (such as XGBoost) excel in feature ranking and variable importance interpretation, they may still obscure the spatiotemporal dependency structures of fire occurrences. Future research should integrate temporal feature extraction or spatial convolutional components from deep learning frameworks to develop hybrid models, thereby enhancing predictive interpretability and spatial generalization capacity. And this study employed a single CMIP6 climate simulation model (BCC-CSM2-MR) for future climate projections, which, while proven reliable for China, may still introduce systematic biases inherent to single-model frameworks. If a multi-model ensemble were adopted, the overall spatial patterns and directional trends of increasing forest fire susceptibility would likely remain robust, as they are primarily driven by consistent warming and drought signals across CMIP6 models. However, the magnitude of projected susceptibility changes may vary among models due to differences in climate sensitivity and precipitation responses, and ensemble approaches would provide a more comprehensive representation of projection uncertainty. Future studies should incorporate multi-model ensemble analyses to reduce structural uncertainty and improve the robustness of fire susceptibility projections. A limitation of the present study was the relatively short historical window of observational climate data, which restricted the ability to fully capture long-term fire-climate dynamics, especially in the face of increasingly frequent extreme weather events. Future work will incorporate high-temporal-resolution remote sensing data and dynamic fuel information to construct time-series-based models, enhancing the simulation of fire processes and adaptation to rapidly changing environments.

Although this study focused on climate-driven fire susceptibility, it did not account for fire responses across vegetation types under future scenarios. Vegetation types vary widely in flammability and combustion characteristics under drought, heat, or low-precipitation stress. In addition, future studies should account for the dynamic evolution of vegetation in response to climate change and human activities. Changes in vegetation structure and fuel composition driven by warming and land-use changes may further influence the spatiotemporal patterns of fire occurrence. Incorporating dynamic vegetation and human activity parameters into future models will enhance the universality and reliability of fire susceptibility projections. Future research will explore ecosystem-specific fire responses to better understand the ecological vulnerability mechanisms under climate change. Our findings highlighted an overall upward trend in fire susceptibility across Yunnan, with spatial patterns shifting from localized to contiguous high-susceptibility zones, especially under SSP5-8.5. Thus, the findings of this study may serve as a reference for regional fire risk management under climate change. While this model emphasized climatic influences, changing human activity and land-use patterns may further shape fire regimes. Therefore, incorporating the synergistic simulations of natural and human drivers may enhance the comprehensiveness and foresight of future fire susceptibility predictions.

5. Conclusions

In summary, this study identified the key determinants and high-susceptibility zones of forest fires in Yunnan during the fire-prevention period and delineated the projected trajectories of fire susceptibility under future climate change. These findings offer a scientific basis for the development of climate-adaptive fire management strategies, early

warning systems, and the efficient allocation of fire prevention resources. Furthermore, the methodology serves as a practical foundation for incorporating land-use change, dynamic fuel data, and cultural intervention mechanisms into future fire prediction frameworks.

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Abbreviations

CMIP6	Coupled Model Intercomparison Project Phase 6
DEM	Digital Elevation Model
ERA5	ECMWF Reanalysis v5
FRP	Fire Radiative Power
GBM	Gradient Boosting Machine
GLC_FCS30	Global Land Cover Fine Classification System (30 m)
MCD14ML	MODIS Collection 6.1 Active Fire Product
MODIS	Moderate Resolution Imaging Spectroradiometer
NDVI	Normalized Difference Vegetation Index
RF	Random Forest
SHAP	Shapley Additive Explanations
SRTM	Shuttle Radar Topography Mission
SSP	Shared Socioeconomic Pathway
SVM	Support Vector Machine
VPD	Vapor Pressure Deficit
XGBoost	Extreme Gradient Boosting
kPa	Kilopascal
RBF	Radial Basis Function

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